

ECON 3201

Econometrics for Economics and Finance

Assignment 1
Due Thursday October 8, 2026

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Instructions

Answer all questions. Please complete your assignment using Quarto. Submit both your Quarto file and pdf output file.

Theory Questions

Q1

Create a personal Quarto website hosted on GitHub pages. For reference, please see the slides for **website** on Nexus. Your website should include the following:

- i. An “Index” or “Home” page. This page should include the following:
 - Your name
 - Your photo (optional)
 - Your credentials
 - An “About Me” section that provides a brief (one paragraph) introduction to yourself. [10 Marks]
- ii. A blog page. This will be the landing site for future assignments [5 Marks]
- iii. Create a blog post and upload the finished PDF for this assignment. [3 Marks]
- iv. Navigation bar with “Home” and “Blog” buttons. The navigation bar is edited in the “_quarto.yml” file. [2 Marks]

Paste your link here

Q2

Write your solutions using LaTeX. Show your work. You may use R to compute the answers.

Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ be a uniformly distributed variable with probability distribution function $f_X(x) = 1/12$.

(a) Check that $f_X(x)$ is a true probability distribution.

$$f_X(x) \text{ is true probability distribution } \sum_{i=1}^{12} f_X(x_i) = 1$$

```
f <- rep(1/12,12)
sum(f)
```

```
[1] 1
```

$$1/12 + 1/12 + 1/12 + 1/12 + \dots + 1/12 = 1$$

(b) Compute the expected value of X .

$$E(X) = \sum_{i=1}^{12} x_i f_X(x)$$

$$E(X) = [(1 \times 1/12) + (2 \times 1/12) + \dots + (12 \times 1/12)] = 6.5$$

```
x <- 1:12
f <- rep(1/12,12)
expected_value <- sum(x*f)
expected_value
```

```
[1] 6.5
```

(c) Compute the variance of X .

$$var(X) = \sum_{i=1}^{12} (x_i - E(X))^2 p_i$$

$$var(X) = [(1 - 6.5)^2 \times 1/12 + (2 - 6.5)^2 \times 1/12 + \dots + (12 - 6.5)^2 \times 1/12] \approx 11.92$$

```
x <- -1:12
f <- rep(1/12, 12)
var <- sum((x - 6.5)^2*f)
var
```

```
[1] 11.91667
```

Q3

Let $X \sim N(4, 5)$. Find the probability of each event:

(a) $Pr(X \leq 6)$

$$Pr(X \leq c_2) = Pr(Z \leq d_2) = \Phi d_2$$

$$Pr(Z \leq 6) = \frac{X - \mu}{\sigma}$$

$$= \frac{6-4}{\sqrt{5}} = 0.8944$$

$$Pr(X \leq 6) = \Phi 0.8944 = 0.8144$$

```
pnorm (6, mean = 4, sd = sqrt(5))
```

```
[1] 0.8144533
```

(b) $Pr(X > 4)$

$$Pr(X > c_1) = Pr(Z > d_1) = 1 - \Phi d_1$$

$$Pr(Z > 4) = \frac{X - \mu}{\sigma}$$

$$= \frac{4-4}{\sqrt{5}} = 0$$

$$Pr(X > 4) = 1 - Pr(X \leq 4)$$

Since $\mu = 4$, Pr of X being greater than 4 is 0.5

$$\therefore, Pr(X > 4) = 1 - 0.5 = 0.5$$

(c) $Pr(-1 \leq X < 1)$

$$Pr(c_1 \leq X < c_2) = Pr(d_1 \leq Z < d_2) = \phi(d_2) - \phi(d_1)$$

$$Pr(-1 \leq X < 1) = Pr(-1 \leq Z < 1) = \phi(1) - \phi(-1) \approx 0.077$$

```
pnorm(1, mean = 4, sd = sqrt(5)) - pnorm(-1, mean = 4, sd = sqrt(5))
```

```
[1] 0.07718259
```

Q4

Let $Z \sim N(0, 1)$.

(a) Compute $Pr(Z \leq 0.83)$

$$Pr(Z \leq d_2) = \frac{X - \mu}{\sigma}$$

$$(Z \leq 0.83) = \frac{0.83 - 0}{1} = \Phi(0.83) \approx 0.797$$

```
pnorm(0.83)
```

```
[1] 0.7967306
```

(b) Compute $Pr(-0.83 \leq Z \leq 0.83)$

$$Pr(d_1 \leq Z \leq d_2) = \Phi(d_2) - \Phi(d_1)$$

$$Pr(-0.83 \leq Z \leq 0.83) = \Phi(0.83) - \Phi(-0.83) \approx 0.593$$

```
pnorm(0.83) - pnorm(-0.83)
```

```
[1] 0.5934612
```

R Exercises

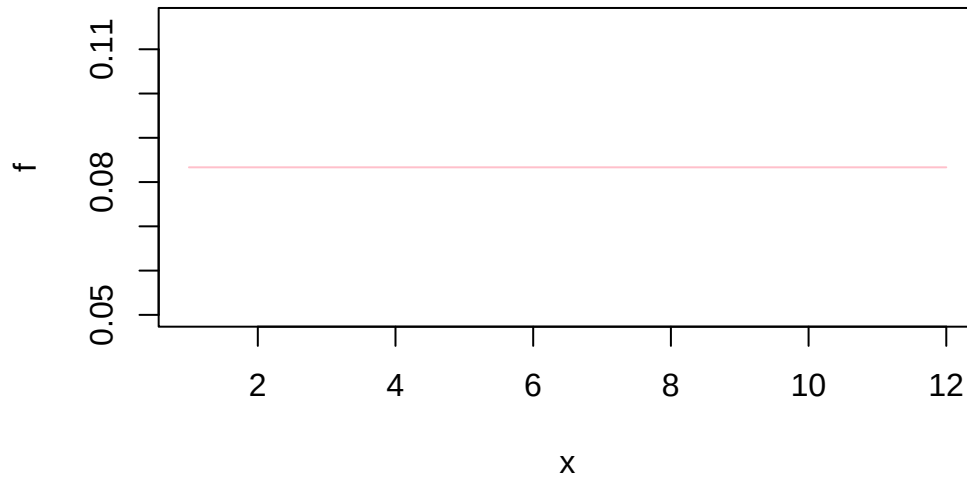
Q5

Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ be a uniformly distributed variable with probability distribution function $f_X(x) = 1/12$.

(a) Plot the probability distribution function $f_X(x)$ in R.

```
x <- 1:12
f <- rep(1/12,12)
plot(x,f,
     type="l",
     main= "Probability Density Function",
     col= "pink", lwd = 1)
```

Probability Density Function

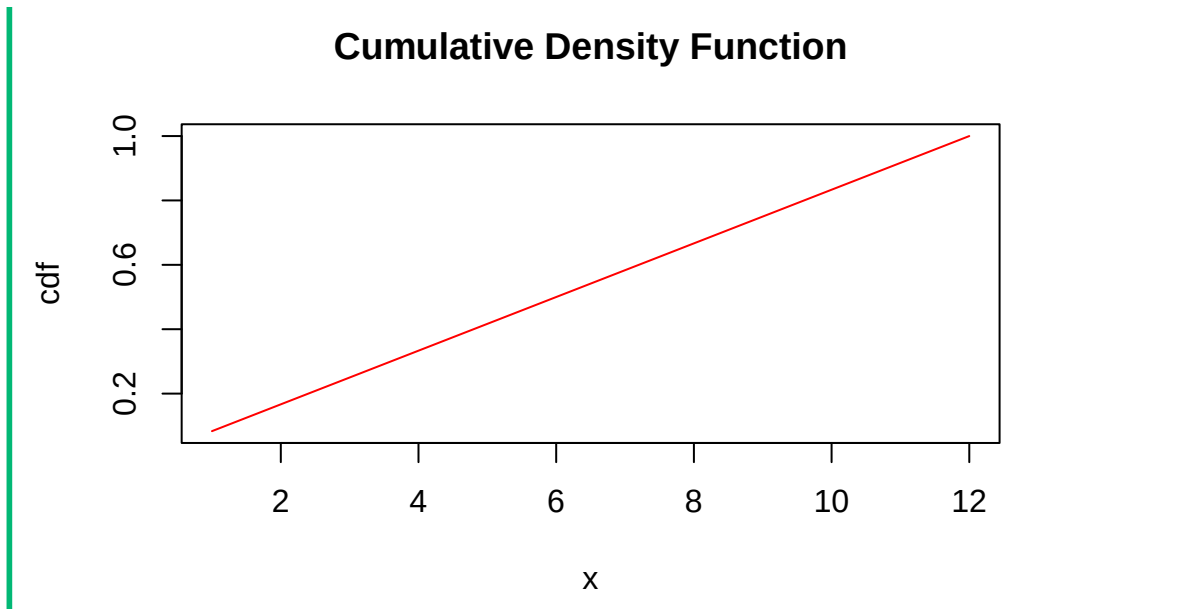


(b) Plot the cumulative probability distribution function $f_X(x)$ in R.

```
x <- 1:12
f <- rep(1/12,12)
cdf <- cumsum(f)
cdf
```

```
[1] 0.08333333 0.16666667 0.25000000 0.33333333 0.41666667 0.50000000
[7] 0.58333333 0.66666667 0.75000000 0.83333333 0.91666667 1.00000000
```

```
plot(x,cdf, type= "l",
     main= "Cumulative Density Function",
     col= "red", lwd = 1)
```



(c) Compute the expected value of X in R.

```
x <- 1:12
f <- rep(1/12,12)
expected_value <- sum(x*f)
expected_value
```

```
[1] 6.5
```

(d) Compute the variance of X in R.

```
x <- 1:12
f <- rep(1/12,12)
expected_value <- 6.5
var <- sum((x-expected_value)^2*f)
var
```

```
[1] 11.91667
```

Q6

The standard normal distribution function is given by

$$f(x) = \frac{e^{-(x^2/2)}}{\sqrt{2\pi}\sigma}$$

- (a) Define this function in R. (Note: π is coded as `pi` and e is coded as `exp`.)

```
f <- function (x) exp(-x^2/2)/sqrt(2*pi)
```

- (b) Integrate this function over the limit $[-\infty, \infty]$ using the `integrate()` command in R. Infinity in R is written as `Inf`. To evaluate the integral, use `integrate()$value`

```
f <- function (x) exp(-x^2/2)/sqrt(2*pi)
integrate(f, -Inf, Inf)$value
```

```
[1] 1
```

$$\int_{-\infty}^{\infty} \times \frac{e^{-x^2/2}}{\sqrt{2\pi}} = 1$$

- (c) Compute the expected value of the standard normal distribution in R and confirm it is equal to 0.

```
f <- function (x) exp(-x^2/2)/sqrt(2*pi)
expected_value_normal <- integrate( function (x) x* f(x), -Inf, Inf)$value
expected_value_normal
```

```
[1] 0
```

$$E[X] = \int_{-\infty}^{\infty} x \left(\frac{e^{-x^2/2}}{\sqrt{2\pi}} \right) dx = 0$$

- (d) Compute the variance of the standard normal distribution in R and confirm it is equal to 1.

```
f <- function (x) exp(-x^2/2)/sqrt(2*pi)
expected_value_normal <- integrate( function (x) x* f(x), -Inf, Inf)$value
var <- integrate( function (x) x^2* f(x), -Inf, Inf)$value
var
```

```
[1] 1
```

$$\text{var}(X) = \int_{-\infty}^{\infty} (x-0)^2 \left(\frac{e^{-x^2/2}}{\sqrt{2\pi}}\right) dx = 1 \quad (x-0)^2 \text{ since } E[X] = 0$$

(e) Compute $Pr(Z \leq 0.83)$

```
f <- function (x) exp(-x^2/2)/sqrt(2*pi)
integrate(f, -Inf, 0.83)$value
```

```
[1] 0.7967306
```

$$\int_{-\infty}^{0.83} \left(\frac{e^{-x^2/2}}{\sqrt{2\pi}}\right) \approx 0.797$$

(f) Compute $Pr(-0.83 \leq Z \leq 0.83)$

```
f <- function (x) exp(-x^2/2)/sqrt(2*pi)
integrate(f, -0.83, 0.83)$value
```

```
[1] 0.5934612
```

$$\int_{-0.83}^{0.83} \left(\frac{e^{-x^2/2}}{\sqrt{2\pi}}\right) \approx 0.593$$

(g) Compute $Pr(-0.83 \leq Z \leq 0.83)$ using the `pnorm()` command and compare your results to (f).

```
f <- function (x) exp(-x^2/2)/sqrt(2*pi)
pnorm(0.83)-pnorm(-0.83)
```

```
[1] 0.5934612
```

$$\text{pnorm}(0.83) - \text{pnorm}(-0.83) \approx 0.593$$

The answer from f is exactly the same as the answer gotten from using the 'pnorm()' command

Q7

Let X denote the prison sentence, in years, for people convicted of auto theft in a particular province in Canada. Suppose the PDF of X is given by:

$$f(X) = \frac{1}{9}x^2, \quad 0 < x < 3.$$

(a) Define $f(x)$ in R.

```
f <- function (x) (1/9)*x^2
```

(b) Integrate $f(x)$ over the interval $[0,3]$.

```
f <- function (x) (1/9)*x^2
integrate(f, 0, 3)$value
```

```
[1] 1
```

$$\int_0^3 \frac{1}{9}x^2 = 1$$

(c) Compute the expected prison sentence.

```
f <- function (x) (1/9)*x^2
expected_prison_sentence <- integrate(function (x) x*f(x), 0, 3)$value
expected_prison_sentence
```

```
[1] 2.25
```

$$\int_0^3 x \times \frac{1}{9}x^2 = 2.25$$

Q8

Let X follow a Student t distribution with M degrees of freedom. Then

$$X = \frac{Z}{\sqrt{W/M}} \sim t_M,$$

where Z is a standard normal variable and W is a chi-squared variable with M degrees of freedom.

(a) On the same figure plot the following:

- (i) Plot the standard normal probability distribution using `curve()`. Specify the range as `[-5,5]` using `xlim`. Name the `ylab` "Density". The main title should be "Densities of t Distribution".
- (ii) Plot the Student t distribution with 5 degrees of freedom using `curve()` and `dt()`. Specify the range as `[-5,5]` using `xlim`. Choose a different colour, i.e. `col=2`. To learn more about `dt`, type `?dt` in your console.
- (iii) Plot the Student t distribution with 15 degrees of freedom using `curve()` and `dt()`. Specify the range as `[-5,5]` using `xlim`. Choose a different colour, i.e. `col=3`. To learn more about `dt`, type `?dt` in your console.
- (iv) Plot the Student t distribution with 25 degrees of freedom using `curve()` and `dt()`. Specify the range as `[-5,5]` using `xlim`. Choose a different colour, i.e. `col=4`. To learn more about `dt`, type `?dt` in your console.
- (v) Include the following legend in you plot:
`legend("topright", c("N(0, 1)", "M=2", "M=4", "M=25"), col = 1:4, lty = c(1, 1, 1, 1))`

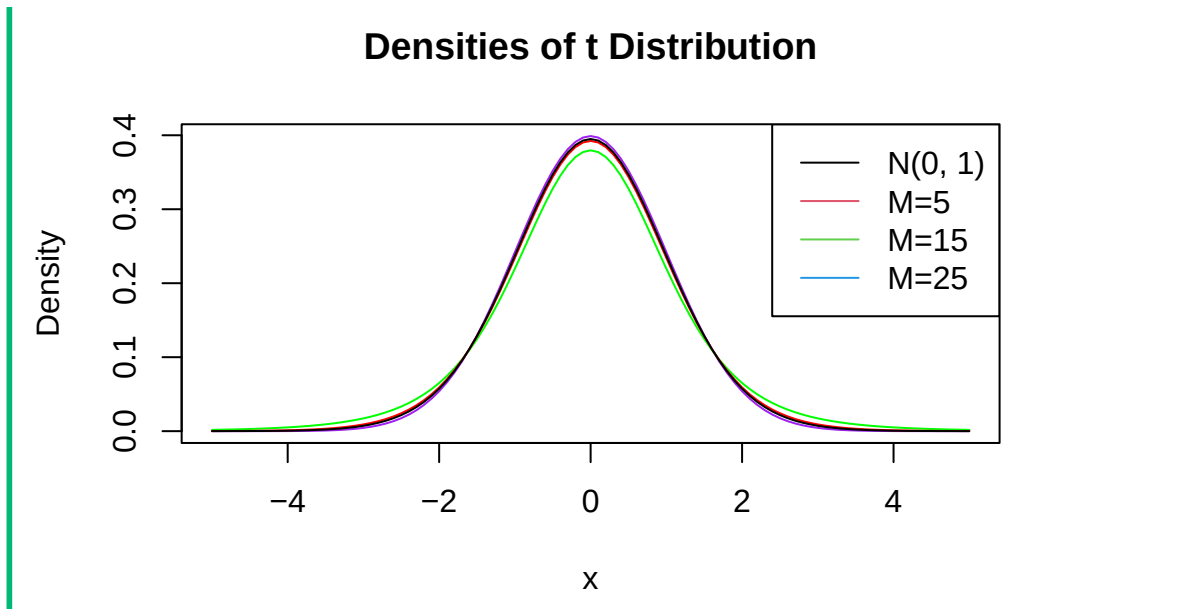
```
curve(dnorm(x), from = -5, to = 5,
      xlim = c(-5,5), ylab = "Density",
      col= "purple",
      main= "Densities of t Distribution")

curve(dt(x, df= 5),
      xlim=c(-5,5), col= "green",
      add= TRUE)

curve(dt(x, df= 15),
      xlim= c(-5,5),col= "red",
      add= TRUE)

curve(dt(x, df= 25),
      xlim= c(-5,5),col= "black",
      add= TRUE)

legend("topright", c("N(0, 1)", "M=5", "M=15", "M=25"), col = 1:4, lty
= c(1, 1, 1, 1))
```



(b) What do you observe as the degrees of freedom increase?

As the df of freedom increase, the t distribution curve gets closer to the normal distribution curve